



Setting BirdNET confidence thresholds: species-specific vs. universal approaches

Sunny Tseng¹ · Dexter P. Hodder² · Ken A. Otter¹

Received: 7 August 2024 / Revised: 12 December 2024 / Accepted: 4 February 2025
© Deutsche Ornithologen-Gesellschaft e.V. 2025

Abstract

BirdNET is widely used in avian acoustic research, providing species predictions alongside confidence values that represent the algorithm's certainty in species identification. Setting thresholds for these confidence values can increase precision (i.e., the percentage of true positives out of all the identified predictions) but may decrease the predictions retained and exclude true positives that fall below the threshold. This study evaluates two methods for setting confidence thresholds using a two-year audio dataset from western Canada, focusing on 19 target species: (1) a universal threshold of 0.7 across all species and (2) species-specific thresholds defined as the minimum confidence required to achieve a precision of at least 0.9. The universal threshold yielded precision ranging from 0.7 to 1.0 across species but retained only $17 \pm 14\%$ (SE) of BirdNET predictions. In contrast, species-specific thresholds ensured precision above 0.9 while retaining $70 \pm 37\%$ (SE) of predictions. Species-specific thresholds varied across species but were generally lower than 0.35. Confidence values associated with BirdNET predictions were found to be species specific, but no clear link was observed between BirdNET's performance and species' song/call complexity, defined as song duration, bandwidth, and number of inflections. Our results confirm that species-specific thresholds offer higher precision and retain more predictions compared to a universal threshold. We provide a step-by-step workflow, including R code, to help researchers define species-specific thresholds that ensure reliable interpretation of BirdNET outputs. Additionally, we discuss how our workflow aligns with and complements previously proposed approaches for setting BirdNET thresholds.

Keywords BirdNET · Performance · Evaluation · Precision · Threshold · Confidence score

Zusammenfassung

Festlegung von BirdNET-Konfidenzschwellen: Artspezifische vs. universelle Ansätze

BirdNET ist in der akustischen Vogelforschung weit verbreitet und liefert Vorhersagen für Arten zusammen mit Konfidenzwerten, welche die Präzision des Algorithmus bei der Artbestimmung darstellen. Die Festlegung von Schwellenwerten für diese Konfidenzwerte kann die Präzision erhöhen (d. h. den Prozentsatz wahrer Positiver unter allen identifizierten Vorhersagen), kann aber auch die Anzahl der identifizierten Vorhersagen verringern und wahre Positive ausschließen, die unter den festgelegten Schwellenwert fallen. In dieser Studie werden zwei Methoden zur Festlegung der Schwellenwerte für Konfidenzintervalle anhand eines zweijährigen Audiodatensatzes aus Westkanada bewertet, wobei der Schwerpunkt auf 19 Zielarten liegt: (1) eine universelle Konfidenzschwelle von 0,7 für alle Arten und (2) artspezifische Konfidenzschwellen, die als Mindestkonfidenz (d. h. kritischer Konfidenzfaktor) definiert sind und erforderlich sind, um eine Präzision von mindestens 0,9 zu erreichen. Der universelle Schwellenwert ergab eine Genauigkeit von 0,7 bis 1,0 für alle Arten, aber nur $17 \pm 14\%$ (SE) der BirdNET-Vorhersagen wurden beibehalten. Im Gegensatz dazu erreichten artspezifische

Communicated by T. S. Osiejuk.

✉ Sunny Tseng
stseng@unbc.ca

¹ Natural Resources and Environmental Studies, University of Northern British Columbia, Prince George, BC, Canada

² John Prince Research Forest, Fort. St. James, BC, Canada

Schwellenwerte eine Genauigkeit von über 0,9 und identifizierten $70 \pm 37\%$ (SE) der Vorhersagen. Die artspezifischen Schwellenwerte variierten je nach Art, lagen aber im Allgemeinen unter 0,35. Es wurde festgestellt, dass die mit den BirdNET-Vorhersagen verbundenen Konfidenzwerte artspezifisch sind, aber es wurde kein klarer Zusammenhang zwischen der BirdNET-Leistung und der Komplexität des Gesangs bzw. der Rufe der Arten, d. h. Dauer des Gesangs, Bandbreite und Anzahl der Beugungen, festgestellt. Unsere Ergebnisse bestätigen, dass artspezifische Schwellenwerte im Vergleich zu einem universellen Schwellenwert eine höhere Präzision bieten und mehr Vorhersagen identifizieren. Wir stellen einen Schritt-für-Schritt-Arbeitsablauf, einschließlich R-Code, zur Verfügung, der Forschenden helfen wird, artspezifische Schwellenwerte zu definieren, die eine zuverlässige Interpretation der BirdNET-Ergebnisse gewährleisten. Darüber hinaus diskutieren wir, wie unser Arbeitsablauf mit zuvor vorgeschlagenen Ansätzen zur Festlegung von BirdNET-Schwellenwerten übereinstimmt und diese ergänzt.

Introduction

Algorithms for bird sound recognition have advanced significantly over recent decades, resulting from collaborations between ornithologists and computer scientists. Key developments include the application of convolutional neural networks for bird sound identification (e.g., Incze et al. 2018; Koh et al. 2019; Tseng et al. 2021), real-time recognition tools (e.g., Kojima et al. 2018; Küçüktopcu et al. 2019), and international competitions in bird sound identification (e.g., Stowell et al. 2019; Kahl et al. 2023). These advancements have led to the creation of commercial software like *Wildlife Acoustics' Kaleidoscope Pro* and open-source tools like the R package *warbleR* (Araya-Salas and Smith-Vidaurre 2017), which automate species identification for both researchers and the broader public. The Cornell Lab of Ornithology's free phone app, Merlin Bird ID, enables real-time species identification and has gained widespread adoption among bird enthusiasts globally. Among these methods, one of the most popular platforms for analyzing such data is BirdNET Analyzer, whose algorithm is available not only as a free phone app (<https://doi.org/https://doi.org/10.1371/journal.pbio.3001670>) but also as open-source code, accessible through a publicly available repository on GitHub (github.com/kahst/BirdNET-Analyzer).

BirdNET Analyzer has emerged as a leading platform for scientific research due to its open-source nature, flexibility, and user-friendly interface. Developed by the K. Lisa Yang Center for Conservation Bioacoustics at Cornell Lab of Ornithology in collaboration with Chemnitz University of Technology, BirdNET was initially released in 2021 with the capacity to identify 984 North American and European bird species (Kahl et al. 2021). Its latest version, BirdNET Analyzer v2.4, extends its capabilities to over 6,000 species worldwide. The algorithm analyzes audio recordings in three-second segments, assigning a 'confidence value' (ranging from 0.1 to 1) to each prediction, where higher values indicate greater certainty. While confidence values reflect

the likelihood of accurate predictions, they do not directly translate into probabilities of true predictions (Pérez-Granados 2023; Wood and Kahl 2024). To address this, the BirdNET team provided guidelines for using confidence values appropriately (Wood and Kahl 2024).

Confidence values can be influenced by various factors, including background noise, signal quality, overlapping vocalizations, and the acoustic characteristics of the species' signals (Symes et al. 2023). For some species, true predictions may occur even at lower confidence values, while for others, high-confidence predictions might still include false positives. Previous studies have noted that the identification performance of BirdNET varies across species (Cole et al. 2022). Such variation can be explored by finding the relationship between BirdNET performance and species vocalization characteristics, such as duration, bandwidth, repertoire size, or mean frequency of songs (Pérez-Granados 2023). This exploration could enhance our understanding of how and why BirdNET performance varies among bird species.

To enhance the reliability of BirdNET predictions, users can set a confidence threshold, which establishes the minimum confidence value required for reporting predictions. While innovative methods have been proposed to bypass thresholds for call density estimation (see Navine et al. 2024), higher thresholds generally increase precision (i.e., the proportion of true positives among identified predictions) but may also exclude true positives with lower confidence values, thereby reducing recall. Retaining true positives (i.e., improving recall) is particularly important in contexts where predictions are used to analyze species' vocal activity patterns over time. Determining an optimal threshold requires balancing high precision with maximizing the number of predictions retained in the dataset. While Wood and Kahl (2024) provided a theoretical framework for setting species-specific thresholds based on probabilities, they did not assess how these species-specific thresholds influence the precision of retained data. Such evaluations are critical for improving

the utility and interpretation of BirdNET outputs (Pérez-Granados 2023; Ware et al. 2023).

In this study, we analyzed a two-year audio dataset from 66 recording sites in northcentral British Columbia, Canada, focusing on 19 target species selected for their diversity in song/call complexity and singing rates. Our objectives were to: (1) compare the effects of a universal BirdNET confidence threshold with species-specific thresholds designed to achieve high precision while retaining a substantial number of predictions, and (2) evaluate whether BirdNET's performance correlates with species-specific song/call complexity. This study provides a detailed workflow, including step-by-step R code, for defining species-specific thresholds and discusses how our method builds on previously proposed approaches for setting BirdNET thresholds. Ultimately, it aims to enhance BirdNET's utility for future research applications.

Methods

Study area

The data were collected at the John Prince Research Forest (54° 27' N, 124° 10' W, 700 m a.s.l), located in the dry sub-boreal spruce biogeoclimatic zone in British Columbia, Canada (Biogeoclimatic Zones of British Columbia 2024). The audio recordings were collected in 66 sites, which were selected to include forest stands of various age classes, tree-species compositions and timber harvesting activities in the region. Adjacent recording sites were at least 2 km apart to ensure independent sampling, accounting for the audio detection radius of the ARUs (Fig. 1).

Each site was equipped with one AudioMoth (Hill et al. 2019), and the data were collected during the breeding seasons (May–July) across two years (2020–2021). The recording window was set from 4 to 7 am daily, aligning with the peak vocal activity of most songbirds. Each AudioMoth was programmed to record actively for one minute, followed by a four-minute period of no recording. This recording cycle was repeated throughout the three-hour recording window,

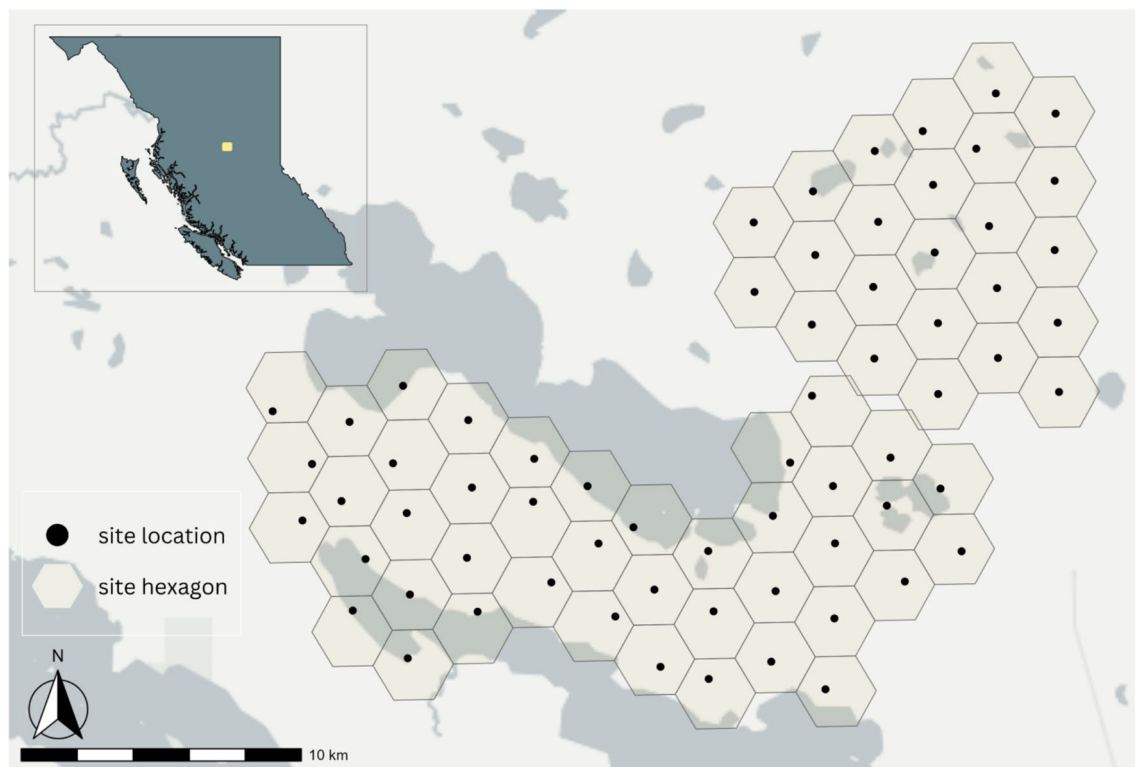


Fig. 1 Locations of the 66 recording sites. All sites are located within or adjacent to John Prince Research Forest, north of Ft. St. James, BC, Canada, with 42 sites within the research forest (lower contiguous cluster) and 24 sites located northeast of the research forest (cluster to upper right of figure). Dots represent the locations of record-

ers, and hexagons were superimposed to ensure spatial independence. Greyed areas on map are lakes and water bodies. All ARUs were initially intended to be positioned at the grid centroids; however, certain site locations deviated slightly due to challenges in reaching the grid centroid

resulting in a total of 36 one-min recordings collected per day, per site. All recordings were digitized at a sampling rate of 48 kHz and stored as 16-bit sound files for subsequent analysis.

Audio data processing

All audio data were processed through BirdNET Lite, a TFLite version of BirdNET. BirdNET enables researchers to specify parameters, enhancing the algorithm's performance. If not explicitly specified, we used the default values for these settings (Table 1). The BirdNET processing was conducted using the high-performance computing lab at the University of Northern British Columbia. Specifically,

we used the *Intel Xeon Silver 4114* cluster, comprising 16 compute nodes, each equipped with 36 *Intel Xeon Gold 6140* core processors. BirdNET's outputs were stored in.csv files.

Target species and song/call complexity

We selected 19 species present in our study area (Table 2), as outlined by one previous study in the same region (Wheelhouse et al. 2022). Among the 19 species, there were six from the warbler family (Parulidae), three from the thrush family (Turdidae), four from the flycatcher family (Tyrannidae), and six selected from a variety of families. The selection aimed to encompass a diverse spectrum of song/call complexity and various singing rates.

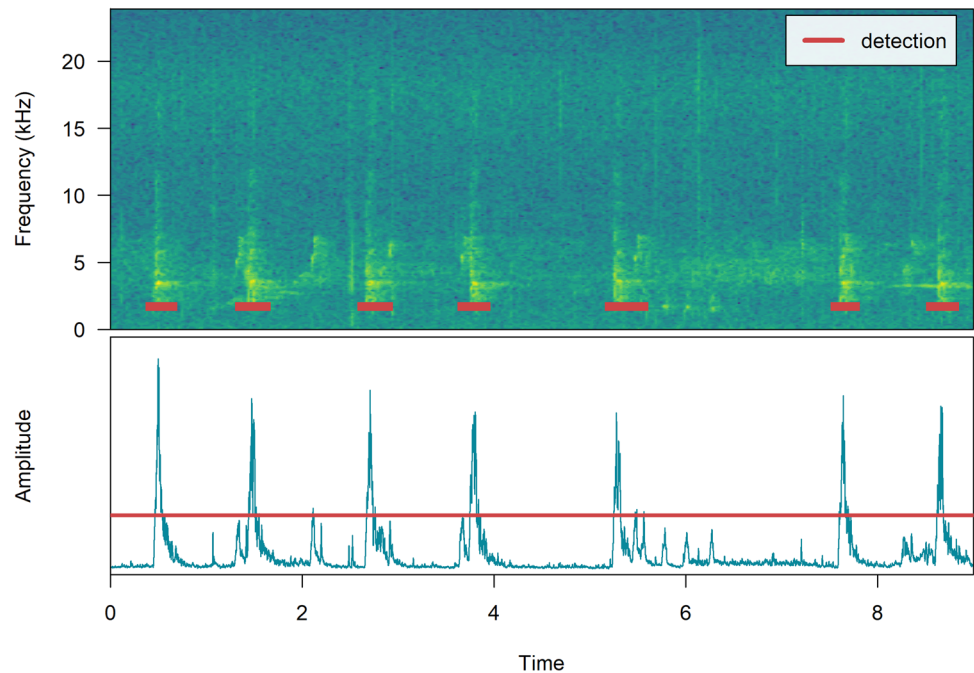
Table 1 Parameters used in this study for the BirdNET analyzer

Arguments	Definition	Values set in this study
i	Path to input file or directory	–
o	Path to output directory. If not specified, the input directory will be used	–
filetype	Filetype of soundscape recordings. Defaults to 'wav'	'wav'
results	Output format of analysis results. Values in ['audacity', 'raven']. Defaults to 'raven'	'raven'
lat	Recording location latitude. Set –1 to ignore	Varies based on site
lon	Recording location longitude. Set –1 to ignore	Varies based on site
week	Week of the year when the recordings were made. Values in [1, 48]. Set –1 to ignore	Varies based on recording date
overlap	Overlap in seconds between extracted spectrograms. Values in [0.0, 2.9]. Defaults to 0	0
spp	Combines probabilities of multiple spectrograms to one prediction. Defaults to 1	1
sensitivity	Sigmoid sensitivity; Higher values result in lower sensitivity. Values in [0.25, 2.0]. Defaults to 1.0	1.0
min_conf	Minimum confidence threshold. Values in [0.01, 0.99]. Defaults to 0.1	0.1

Table 2 Target species in this study

Family	Common name	Scientific name
New World Warblers (Parulidae)	Northern Waterthrush	<i>Parkesia noveboracensis</i>
	Orange-crowned Warbler	<i>Leiothlypis celata</i>
	Tennessee Warbler	<i>Leiothlypis peregrina</i>
	MacGillivray's Warbler	<i>Geothlypis tolmiei</i>
	Wilson's Warbler	<i>Cardellina pusilla</i>
	Myrtle Warbler	<i>Setophaga coronata</i>
Thrush (Turdidae)	American Robin	<i>Turdus migratorius</i>
	Swainson's Thrush	<i>Catharus ustulatus</i>
	Varied Thrush	<i>Ixoreus naevius</i>
Tyrant Flycatcher (Tyrannidae)	Hammond's Flycatcher	<i>Empidonax hammondii</i>
	Olive-sided Flycatcher	<i>Contopus cooperi</i>
	Western Wood-Pewee	<i>Contopus sordidulus</i>
	Western Flycatcher	<i>Empidonax difficilis</i>
Woodpeckers (Picidae)	Northern Flicker	<i>Colaptes auratus</i>
Wrens (Troglodytidae)	Pacific Wren	<i>Troglodytes pacificus</i>
Kinglets (Regulidae)	Ruby-crowned Kinglet	<i>Corthylio calendula</i>
Nuthatches (Sittidae)	Red-breasted Nuthatch	<i>Sitta canadensis</i>
Treecreepers (Certhiidae)	Brown Creeper	<i>Certhia americana</i>
New World Sparrows (Passerellidae)	White-throated Sparrow	<i>Zonotrichia albicollis</i>

Fig. 2 Spectrogram illustrating the detection of song/call signals from a sound recording using the *energy_detector()* function in the *ohun* package. The example recording is from an American Robin. For each signal, the duration (s), bandwidth (Hz), and number of inflections were calculated and averaged throughout the recording. A total of 50 nine-second recordings were selected and analyzed for each target species to quantify song/call complexity



We quantified the song/call complexity for each target species using signal duration (s), bandwidth (Hz), and number of inflections (defined as the average number of frequency inflections in a song/call signal). The value for each metric was calculated from 50 nine-second recordings for each target species. First, we assessed the song/call signals within a recording using the *energy_detector()* function in the *ohun* package (Araya-Salas et al. 2023), which detected the timing and the duration of the signal (Fig. 2). Then we used the package *warbleR* (Araya-Salas and Smith-Vidaurre 2017) to calculate the bandwidth and the number of inflections for each song/call signal. Specifically, the bandwidth was calculated using the *freq_range()* function, and the number of inflections was calculated using the *freq_ts()* and the *inflections()* functions. We selected multiple metrics to quantify complexity because single metrics might not be sufficient when comparing complexity between species (Benedict and Najar 2019).

Validation dataset

For each of the 19 target species used in the full analysis, we employed a stratified random sampling approach, selecting ten recording segments from each BirdNET confidence value class with a 0.05 incremental grouping. In other words, for each species, we randomly selected ten three-second recording segments which BirdNET had identified as containing the target species with confidence values ranging from 0.1 to 0.15; another ten recording segments with confidence values ranging from 0.15 to 0.20, and so forth through 0.95–1 confidence values. This process yielded a validation

dataset consisting of 180 recording segments for each target species. The application of stratified sampling ensured a balanced validation dataset, mitigating potential bias introduced by the varying number of predictions assessed within each confidence class.

The decision to include 180 predictions per target species was informed by a pilot study on a subset of species, which examined how the number of three-second recording segments influences BirdNET performance evaluation. In this pilot study, we systematically chose 50, 25, 10, and 5 recording segments from each confidence class for three species: Varied Thrush (*Ixoreus naevius*), Olive-sided Flycatcher (*Contopus cooperi*), and Myrtle Warbler (*Setophaga coronata*). While a larger number of segments in the validation dataset intuitively enhances result accuracy, practical constraints, particularly the manual effort required for validating multiple species, must also be considered. Based on this analysis (presented at the beginning of the Results section), we selected ten recording segments per confidence value class (i.e., 180 recording segments for each target species) as achieving a balance between validation effort and quality.

Calibration curves—linking confidence value to true positive rate

All 180 sampled recording segments were verified through both audio and visual inspection of spectrograms by the first author. Each audio segment was manually reviewed to confirm the presence or absence of the target species as reported by BirdNET. Predictions confirmed as present

were classified as true positives (1), while those confirmed as absent were classified as false positives (0). We then fitted a generalized linear model with a binomial distribution to generate a calibration curve, using BirdNET confidence values as the predictor variable and true/false positive classifications as the response variable. Logistic regression, as a method to model the relationship between BirdNET confidence values and the true positive rate, was also proposed by Wood and Kahl (2024). The true positive rate was calculated by back-transforming the model predictions.

Confidence threshold setting

The steps below outline the calculation for precision (i.e., the percentage of true positives out of all the identified predictions) and the number of remaining predictions under varying thresholds for single species. This iterative procedure was applied to all 19 target species within our study:

1. No. of true positives: for each possible confidence value, we multiplied the true positive rate (predicted from the calibration curve) by the number of predictions with that specific confidence value:

$$N_{TP_i} = TPR_i \times N_i$$

2. No. of false positives: for each possible confidence value, multiply $(1 - \text{true positive rate})$ by the number of predictions with that specific confidence value:

$$N_{FP_i} = (1 - TPR_i) \times N_i$$

3. Number of remaining predictions: the sum of true positives and false positives whose confidence value is higher than the confidence threshold:

$$\text{No. of remaining detections} = \sum_{i=T}^1 N_{TP_i} + \sum_{i=T}^1 N_{FP_i}$$

4. Precision: the sum of all true positives divided by the number of remaining predictions:

precision (i.e., proportion of true positives)

$$= \sum_{i=T}^1 N_{TP_i} / \sum_{i=T}^1 N_{TP_i} + \sum_{i=T}^1 N_{FP_i}$$

where: i represents all possible confidence values, ranging from 0.1 to 1. N_{TP_i} is the number of true positives with confidence i . N_{FP_i} is the number of false positives with confidence i . TPR_i is the true positive rate with confidence i , derived from calibration curve. N_i is the total number of predictions with confidence i . T is the confidence threshold, with potential values ranging from 0.1 to 1.

Based on the calculations, we examined two different scenarios for selecting confidence thresholds and how these

scenarios influenced precision and the number of remaining predictions: (1) universal threshold: a general high threshold of 0.7, as summarized in Table 2 in Pérez-Granados (2023), to ensure that retained predictions were most likely true positives; and (2) species-specific threshold: defined as the minimum confidence threshold the user can apply to achieve a precision of at least 0.9. This method allows us to specify that at least 90% of the predictions remaining in the dataset are true positives. This target precision (0.9) is set for demonstration purposes in this study, future users can define their own desired precision as the criteria.

Results

Linking confidence value to true positive rate

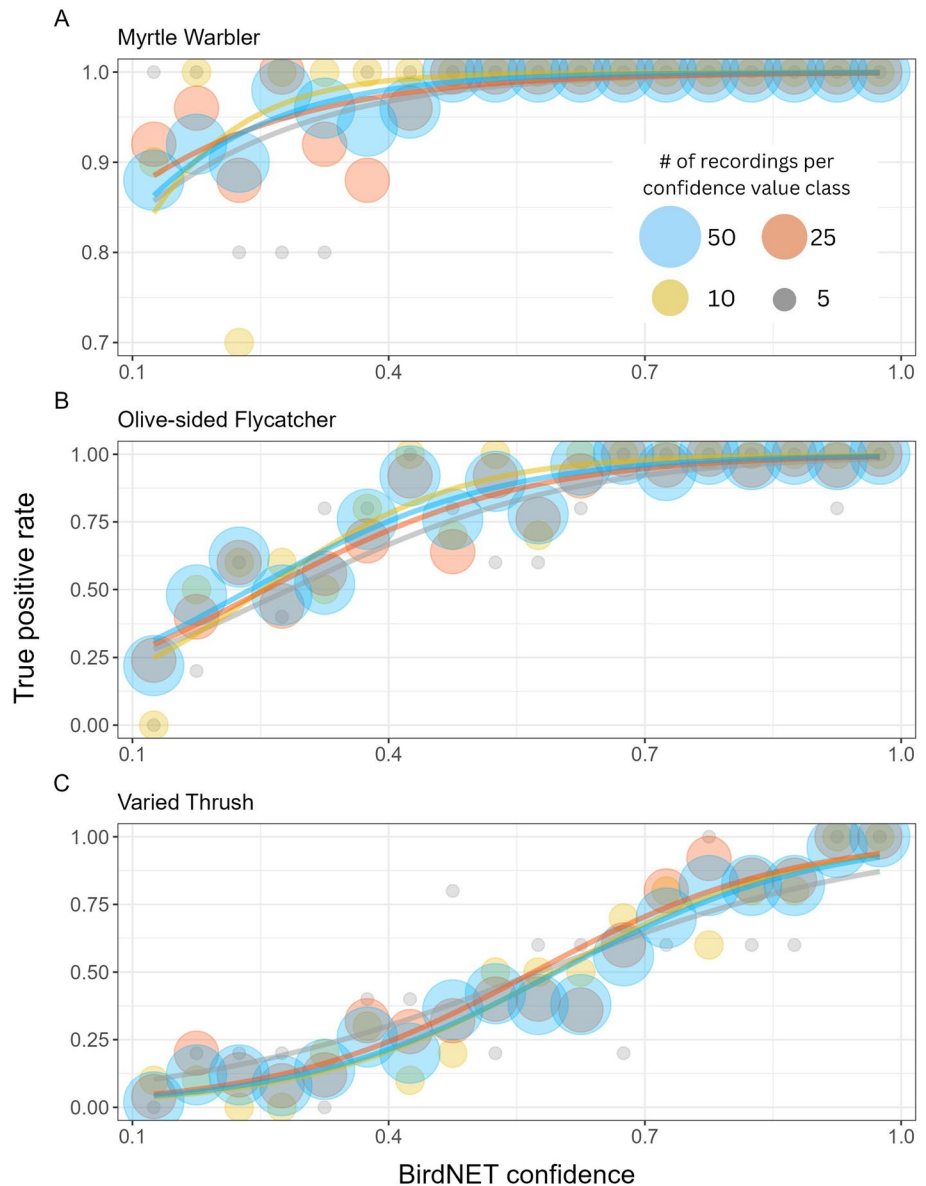
There was a positive relationship between BirdNET confidence and the true positive rate regardless of species, or whether 5–50 recording segments were selected (Fig. 3). Calibration curves derived from datasets comprising 50, 25, and 10 recording segments per confidence class exhibited comparable trends. However, the curve generated from five recording segments per confidence class diverged to a greater extent from the remaining calibration curves (see the gray curve in Fig. 3c); therefore, to balance the effort invested in manual validation, we selected ten segments per confidence class (180 segments per target species) in the validation dataset in this study.

We established calibration curves for our target species, employing a standardized sample size of ten recording segments per confidence class (180 segments per target species; Fig. 4). Species such as the Myrtle Warbler (green), Red-breasted Nuthatch (yellow), and Pacific Wren (orange) attained high true positive rates even at low BirdNET confidence, suggesting that predictions are likely to be accurate despite low confidence values. Conversely, true positive rates exhibited gradual increase in relation to confidence values for species like Varied Thrush (purple) and Brown Creeper (blue), indicating that only predictions with high confidence were likely to be accurate. Species such as Olive-sided Flycatcher (pink) and Northern Flicker (dark green) fell between these two groups.

Impact of confidence threshold setting

With a universal confidence threshold of 0.7, the precision was consistently high, ranging from 0.7 to 1.0 (Table 3). However, applying this universal threshold resulted in lower number of BirdNET predictions being retained (blue arrows in Fig. 5). Specifically, only $17 \pm 14\%$ (SE) of BirdNET

Fig. 3 Calibration curves to estimate appropriate number of recordings in the validation dataset for (A) Myrtle Warbler, (B) Olive-sided Flycatcher, and (C) Varied Thrush. Curves were generated using 5 (grey), 10 (yellow), 25 (orange), and 50 (blue) recording segments per BirdNET confidence value class (color figure online)



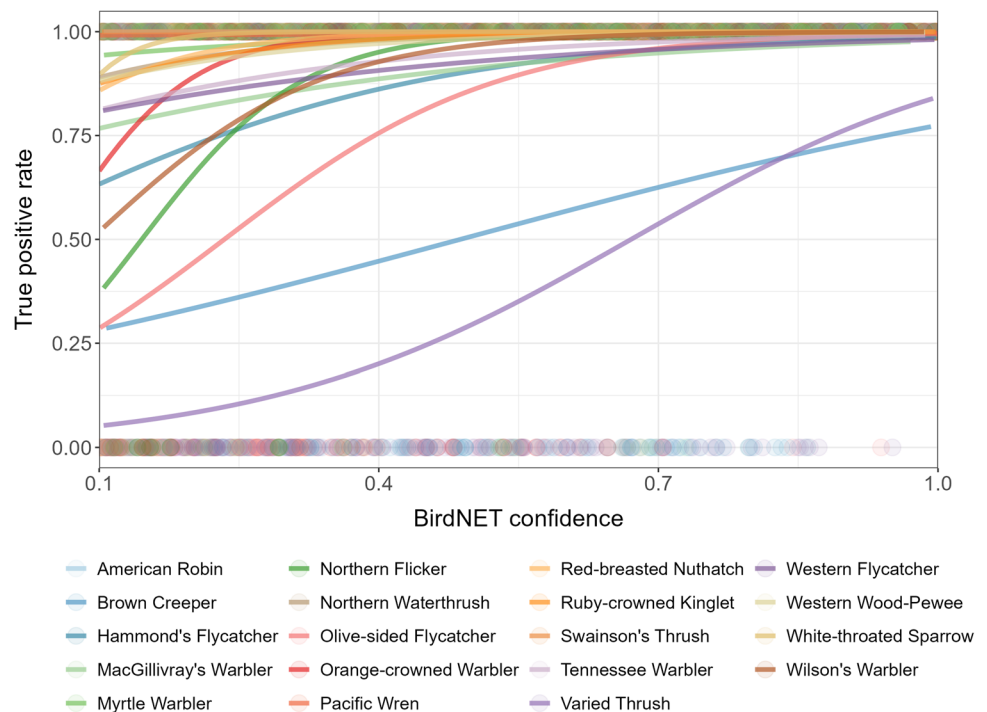
predictions were retained in the dataset when applying a universal threshold of 0.7.

Conversely, the species-specific threshold achieved precisions higher than 0.9 (Table 3), while also retained higher number of predictions in the dataset (red arrows in Fig. 5). Overall, $70 \pm 37\%$ (SE) of BirdNET predictions were retained in the dataset when applying species-specific thresholds. These species-specific thresholds were lower than 0.35 for most of our target species, with only two of the 19 target species, Brown Creeper (0.95) and Varied Thrush (0.95), requiring higher thresholds to reach similar level of precision (Table 3).

Song/call complexity vs BirdNET performance

We did not find a clear link between BirdNET's performance (defined as the threshold needed to achieve at least 0.9 precision, as shown in Table 3, species-specific threshold column) and song/call complexity (defined as signal duration, bandwidth, and the number of inflections) among the species we tested (Fig. 6). Examples of species for which BirdNET performs poorly (i.e., needing a higher threshold to achieve 0.9 precision) included Varied Thrush and Brown Creeper. Varied Thrush had the narrowest bandwidth among all species measured but has intermediate signal duration and number of

Fig. 4 Calibration curves for the 19 target species, comparing true positive rate versus BirdNET confidence. The curve was based on logistic regression with true positive (1) or false positives (0) as response variables, and BirdNET confidence values as the predictor variable. The true positive rate was calculated via back-transformation of the model predictions



inflections (yellow highlight in Fig. 6). Similarly, the Brown Creeper exhibits intermediate values for signal duration, bandwidth, and number of inflections (yellow highlight in Fig. 6). In comparison, the White-throated Sparrow had lower complexity values for all three metrics but required low BirdNET confidence threshold to reach high precision (green highlight in Fig. 6).

Discussion

Overview of BirdNET performance and thresholds

Our results suggested setting species-specific confidence thresholds while using BirdNET may be a better option to balance between precision and the number of predictions retained. For species that BirdNET had high success at identifying, such as the Myrtle Warbler (Fig. 5a), users could set a low confidence threshold (e.g., 0.1). Conversely, for species like the Olive-sided Flycatcher (Fig. 5b), a moderate confidence threshold (e.g., 0.34) would be needed to achieve a balance between retaining predictions in the dataset while maintaining a high true-positive rate. In contrast, species like the Brown Creeper (Fig. 5c) and Varied Thrush (Fig. 5d) would require setting a higher confidence threshold (e.g., 0.95) to avoid inclusion of false positives. Although this threshold retains only about 1% of the original predictions

for these latter species, it also ensures a higher certainty of true positive predictions in the final dataset.

We did not find a correlation between song/call complexity (defined by duration, bandwidth, and number of inflections) and BirdNET performance (defined as the threshold needed to achieve at least 0.9 precision) among the 19 target species. This suggests that BirdNET may not necessarily perform better at detecting species with higher vocal complexity. However, this study considered only a limited set of song and call metrics, focusing on the 19 target species in our analysis. Many other species present in recordings from our study site remain unexamined. Future research could explore additional complexity metrics, such as repertoire: the number of total song types (e.g., Weiss et al. 2014), delivery: the number of songs recorded over total recording time (e.g., Opaev 2016), switching rate: the proportion of songs which represent a type switch from the preceding song (e.g., Hill et al. 2018), and variety: short-term song diversity (e.g., dos Santos et al. 2018). Given the low correlation between these metrics (Benedict and Najar 2019), multivariate approaches could help identify the primary factors influencing BirdNET performance. Expanding analyses to include a broader range of species and examining the influence of specific song and call types within species could yield deeper insights into BirdNET's performance.

We did find that misclassifications appeared to be due to overlapping acoustic features between species. During

Table 3 Statistics for each target species, including the total number of detections, thresholds, corresponding precision, and the proportion of detections retained after applying the thresholds

Common name	Scientific name	No. of detections	Universal			Species-specific		
			Threshold	Precision	Proportion retained (%)	Threshold	Precision	Proportion retained (%)
American Robin	<i>Turdus migratorius</i>	21,284	0.7	1.00	6	0.10	1.00	100
Swainson's Thrush	<i>Catharus ustulatus</i>	376,293	0.7	1.00	32	0.10	1.00	100
Pacific Wren	<i>Troglodytes pacificus</i>	45,232	0.7	1.00	39	0.10	0.99	100
White-throated Sparrow	<i>Zonotrichia albicollis</i>	39,869	0.7	1.00	7	0.10	0.98	100
Northern Waterthrush	<i>Parkesia noveboracensis</i>	89,340	0.7	1.00	31	0.10	0.97	100
Red-breasted Nuthatch	<i>Sitta canadensis</i>	9007	0.7	1.00	35	0.10	0.97	100
Myrtle Warbler	<i>Setophaga coronata</i>	30,418	0.7	1.00	2	0.10	0.96	100
Ruby-crowned Kinglet	<i>Corthylio calendula</i>	13,005	0.7	1.00	13	0.10	0.94	100
Western Wood-Pewee	<i>Contopus sordidulus</i>	4414	0.7	1.00	13	0.10	0.94	100
Tennessee Warbler	<i>Leiothlypis peregrina</i>	27,162	0.7	0.99	45	0.10	0.94	100
Orange-crowned Warbler	<i>Leiothlypis celata</i>	12,636	0.7	1.00	5	0.15	0.93	61
Northern Flicker	<i>Colaptes auratus</i>	853	0.7	1.00	17	0.18	0.90	48
Hammond's Flycatcher	<i>Empidonax hammondi</i>	92,695	0.7	0.98	31	0.19	0.90	75
Western Flycatcher	<i>Empidonax difficilis</i>	11,698	0.7	0.97	6	0.19	0.90	39
Wilson's Warbler	<i>Cardellina pusilla</i>	3484	0.7	1.00	4	0.22	0.90	33
MacGillivray's Warbler	<i>Geothlypis tolmiei</i>	24,083	0.7	0.96	7	0.27	0.90	31
Olive-sided Flycatcher	<i>Contopus cooperi</i>	11,388	0.7	0.99	18	0.34	0.90	34
Varied Thrush	<i>Ixoreus naevius</i>	24,525	0.7	0.71	7	0.95 ^a	0.82 ^a	1 ^a
Brown Creeper	<i>Certhia americana</i>	12,403	0.7	0.70	5	0.95 ^a	0.76 ^a	1 ^a

^aVaried Thrush and Brown Creeper did not achieve a precision of 0.9, even at highest thresholds. The reported values are based on thresholds set to 0.95

Precision varied across species when applying a universal BirdNET threshold of 0.7, with an average of 17% of detections retained from the unfiltered dataset. In contrast, species-specific thresholds, defined as the minimum threshold to achieve a precision greater than 0.9, retained an average of 70% of detections. This table serves as an example, and researchers can adjust the precision criteria for species-specific thresholds based on their specific objectives

validation, we found that Swainson's Thrush calls were frequently misidentified as Olive-sided Flycatcher. The call of Olive-sided Flycatcher has a similar tonal quality and frequency range as notes in the Swainson's Thrush call. We also found that most Varied Thrush predictions with low BirdNET confidence values were actually White-throated Sparrow songs. This may be due to the similar monotone whistles and overlapping frequency ranges shared by the two species' songs. Interestingly, White-throated Sparrow predictions were mostly accurate and rarely misidentified as Varied Thrush. This might be because BirdNET had a broader training dataset for White-throated Sparrow. For future research, we recommended evaluating how the number, and the variety, of the training dataset used to build the learning algorithm could influence the BirdNET performance for each species.

Applications of two types of thresholds

In practice, while species-specific thresholds have been strongly recommended (Fairbairn et al. 2024; Wood and Kahl 2024, and this study), a universal threshold may still have utility for exploratory analyses, such as preliminary assessments of species richness in large-scale biodiversity surveys (Ware et al. 2023). However, using a universal threshold can result in both false positives (species predicted by BirdNET but not actually present) and false negatives (species present at the site but not predicted by BirdNET). As a result, it is difficult to recommend a 'precise' universal threshold that would reliably reflect species richness. Manual validation of predictions still remains a crucial and necessary step even when employing universal thresholds in species richness studies.

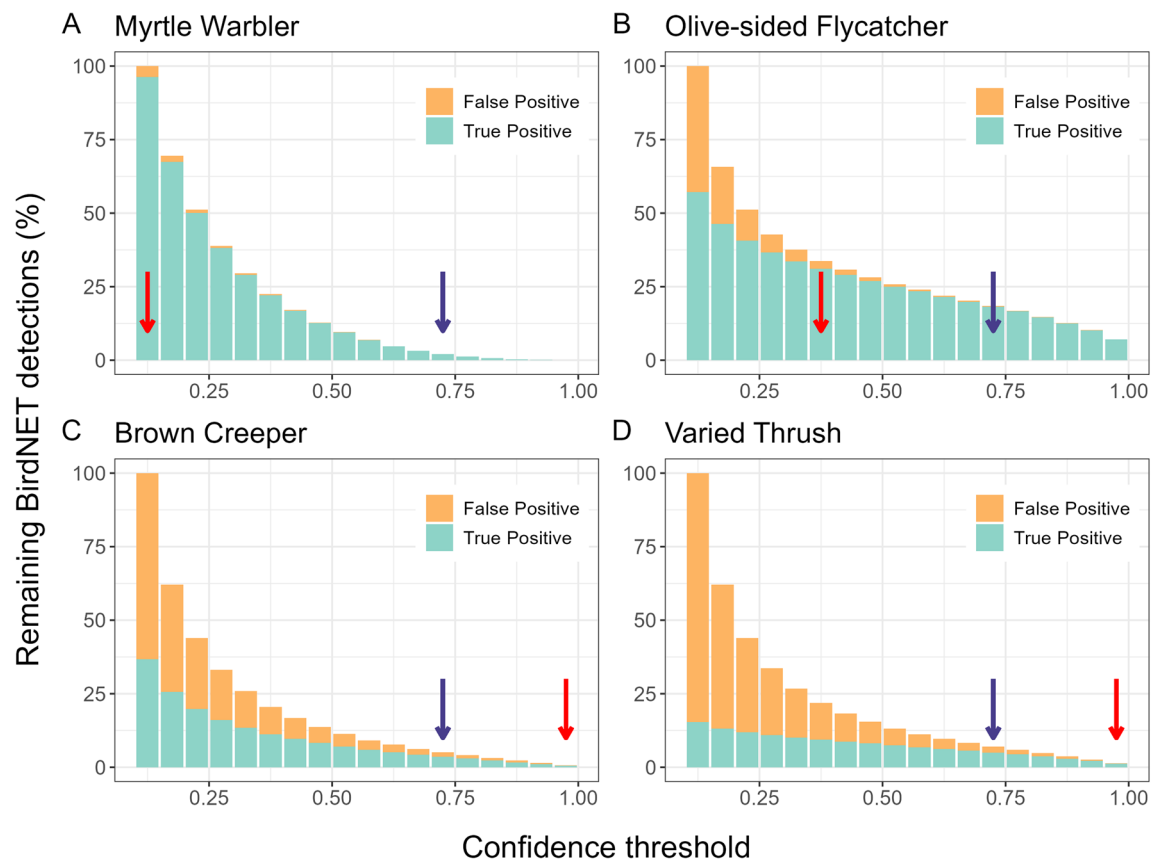


Fig. 5 The number of detections identified by BirdNET, with true positives (green) or false positives (orange), after applying various confidence thresholds, ranging from 0.1 to 0.95, with 0.05 increment. We show results for (A) Myrtle Warbler, (B) Olive-sided Flycatcher, (C) Brown Creeper, and (D) Varied Thrush. Blue arrows represented the universal threshold of 0.7 and the red arrows represented the

species-specific thresholds for each species, defined as the minimum confidence threshold achieving a precision of at least 0.9. Note that Brown Creeper and Varied Thrush did not achieve 0.9 precision, even with the highest threshold. For these species, the species-specific threshold was set to 0.95 (color figure online)

For more detailed investigations, such as analyzing species vocal activity patterns or tracking rare species, species-specific thresholds are essential. These thresholds help maximize true-positive predictions, capturing a more complete vocal census (Table 3). Using species-specific thresholds prevents biased conclusions that could arise from confidence values (within species) that vary across either locations or time periods, providing a more balanced and realistic representation of vocal activity.

Workflow to define species-specific threshold

The species-specific thresholds calculated in this study (Table 3) are not transferable to different regions, even for the same species, as they may be influenced by factors such as recording equipment, ambient noise, environmental conditions, and the local species composition. To support

future studies, we summarized our proposed steps for defining species-specific BirdNET confidence thresholds (Fig. 7) and provided the corresponding R code and example datasets as electronic supplementary materials. The R code also includes a developed R Shiny App for BirdNET validation, which integrates both audio and spectrographic options for assessing predictions.

While some aspects of our proposed methods align with the approaches outlined by Wood and Kahl (2024) in their ‘converting scores to probabilities’ section, there are several important distinctions. First, during validation, we recommend stratified sampling across confidence values to ensure representation of both low- and high-confidence predictions, rather than the simple random sampling. The stratified sampling approach is particularly advantageous when confidence value distributions are non-uniform, a common scenario in BirdNET analyses.

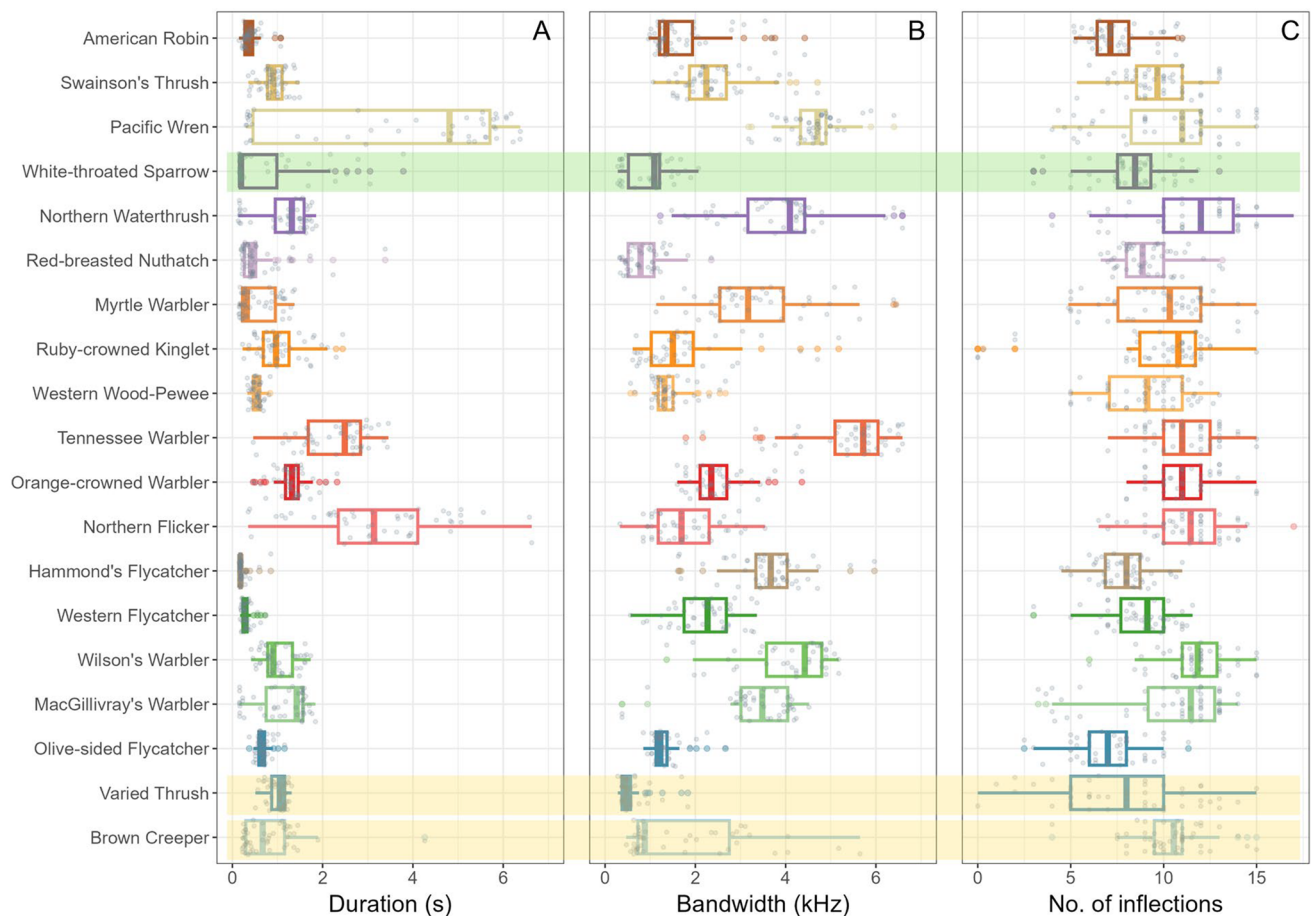


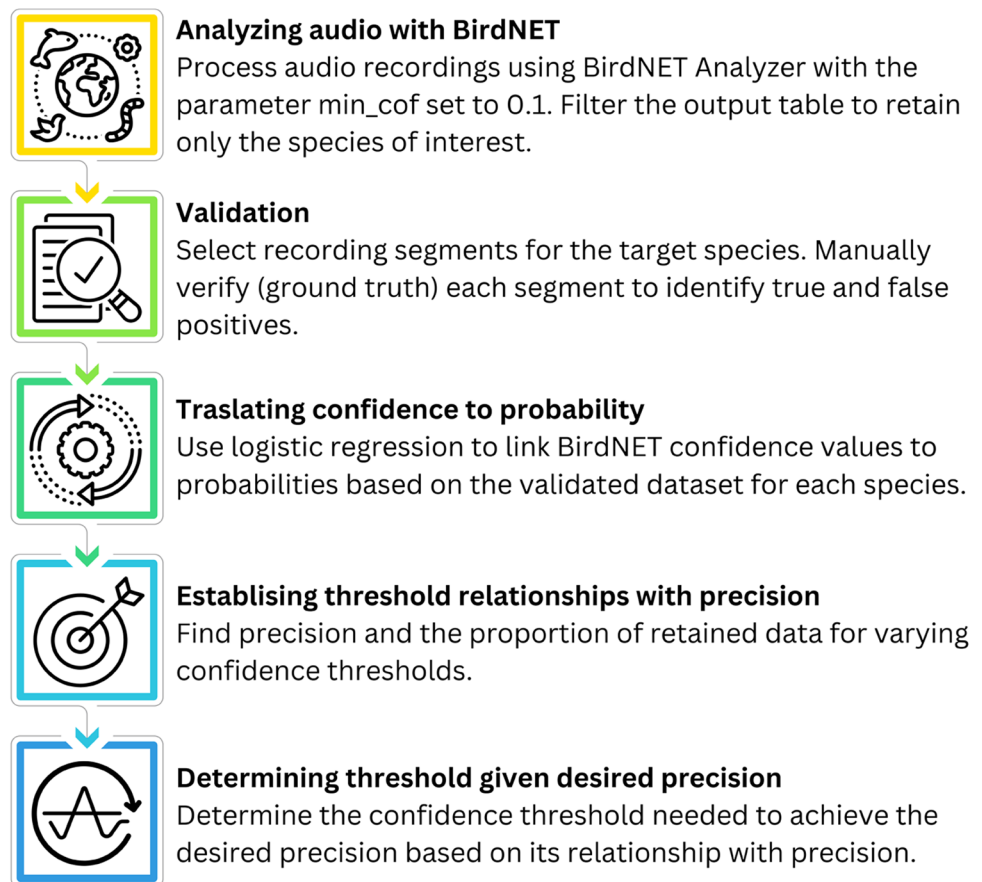
Fig. 6 Vocal complexity of the 19 target species ranked by the BirdNET's performance (defined as the threshold needed to achieve 0.9 precision, as shown in Table 3). Vocal complexity is defined by (A) duration of the song/call in seconds, (B) bandwidth of the song/call in kHz, and (C) the number of inflections of the signals. Boxes show quartiles and median of each metrics. Metrics were measured from

50 recordings for each species. BirdNET performed poorly for species highlighted in yellow and performed well for species highlighted in green. The duration of the signals was defined by the *every_detector()* function in the *ohun* package. The bandwidth and the number of inflections were extracted by the *freq_range()* and the *inflections()* functions in the *warbleR* package, respectively (color figure online)

The number of validation segments required may vary depending on the spatiotemporal scale and the diversity of vocalizations in the study. In our study, we validated 180 segments per species; however, for projects with greater vocalization variability, we recommend testing how segment numbers influence the calibration curve, as demonstrated in the Results section '[Linking confidence value to true positive rate](#)'. For large-scale or highly diverse datasets, rather than selecting more segments and determining thresholds per species, users may consider defining thresholds at finer scales, such as by species and site, to better capture variability.

Second, while both methods involve deriving thresholds from logistic regression models, our approach emphasizes precision, assessing the proportion of retained predictions that are true positives. In contrast, Wood and Kahl's (2024) method focuses on the probability of individual predictions being true positives. These differences lead to different threshold values: our method identifies the confidence threshold at which retained predictions achieve a desired precision, whereas theirs determines the confidence value where the predicted probability equals a given value. We encourage users to evaluate the objectives and select the appropriate method for defining thresholds to enhance the utility of BirdNET.

Fig. 7 Workflow for defining species-specific thresholds. For detailed method descriptions and implementation steps, refer to the R code provided in the supplemental material



Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10336-025-02260-w>.

Acknowledgements We thank Tl'azt'en Nation, Binche Whut'en, and Nak'azdli Whut'en for supporting this work on their territories and their collaboration in the data collection. We thank Shannon Crowley, Lauren Wheelhouse, Morgan Conrad, Steven Murdock, and Amanda Carriere for helping collect ARU data.

Funding This project was funded by Natural Science and Engineering Research Council of Canada (Discovery Grant to KAO, CGS-D to ST), Habitat Conservation Trust Foundation (DPH), John Prince Research Forest (DPH), and Mitacs Accelerate (ST & KAO).

Data availability A step-by-step tutorial, R code, and example datasets are available in the electronic supplementary material. All analysis R scripts used in this study are deposited in a GitHub repository (https://github.com/SunnyTseng/thesis_aru_BirdNET_evaluation).

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical statement Data were collected under approval of the UNBC Animal Care and Use Committee (protocol—2022-01).

References

- Araya-Salas M, Smith-Vidaurre G (2017) warbleR: An R package to streamline analysis of animal acoustic signals. *Methods Ecol Eval* 8:184–191. <https://doi.org/10.1111/2041-210X.12624>
- Araya-Salas M, Smith-Vidaurre G, Chaverri G, Brenes JC, Chirino F, Elizondo-Calvo J, Rico-Guevara A (2023) Ohun: an R package for diagnosing and optimizing automatic sound event detection. *Methods Ecol Eval* 14:2259–2271. <https://doi.org/10.1111/2041-210X.14170>
- Benedict L, Najar NA (2019) Are commonly used metrics of bird song complexity concordant? *Auk* 136:uky008. <https://doi.org/10.1093/auk/uky008>
- Cole JS, Michel NL, Emerson SA, Siegel RB (2022) Automated bird sound classifications of long-duration recordings produce occupancy model outputs similar to manually annotated data. *Ornithol Appl* 124:duac003. <https://doi.org/10.1093/ornithapp/duac003>
- dos Santos EB, Llambías PE, Rendall D (2018) Male song diversity and its relation to breeding success in southern House Wrens *Troglodytes aedon* Chilensis. *J Avian Biol* 49:e01606. <https://doi.org/10.1111/jav.01606>
- Fairbairn AJ, Burmeister J-S, Weisser WW, Meyer ST (2024) BirdNET is as good as experts for acoustic bird monitoring in a European city. *bioRxiv*: preprint. <https://doi.org/10.1101/2024.09.17.613451>
- Hill SD, Brunton DH, Anderson MG, Ji W (2018) Fighting talk: complex song elicits more aggressive responses in a vocally complex songbird. *Ibis* 160:257–268. <https://doi.org/10.1111/ibi.12542>

- Hill AP, Prince P, Snaddon JL, Doncaster CP, Rogers A (2019) Audio-Moth: a low-cost acoustic device for monitoring biodiversity and the environment. *HardwareX* 6:e00073. <https://doi.org/10.1016/j.ohx.2019.e00073>
- Incze Á, Jancsó H-B, Szilágyi Z, Farkas A, Sulyok C (2018) Bird sound recognition using a convolutional neural network. In: *Proceedings of 2018 IEEE 16th international symposium on intelligent systems and informatics (SISY)*. <https://doi.org/10.1109/SISY.2018.8524677>
- Kahl S, Wood CM, Eibl M, Klinck H (2021) BirdNET: a deep learning solution for avian diversity monitoring. *Ecol Inform* 61:101236. <https://doi.org/10.1016/j.ecoinf.2021.101236>
- Kahl S, Denton T, Klinck H, Reers H, Cherutich F, Glotin H, Goëau H, Vellinga W-P, Planqué R, Joly A, Yang K (2023) Overview of BirdCLEF 2023: automated bird species identification in eastern Africa 4.0 international (CC BY 4.0). In: *CLEF 2023: conference and labs of the evaluation forum*, September 18–21, Thessaloniki, Greece
- Koh C-Y, Chang J-Y, Tai C-L, Huang D-Y, Hsieh H-H, Liu Y-W (2019) Bird sound classification using convolutional neural networks (CC BY 4.0). In: *2018 IEEE 16th international symposium on intelligent systems and informatics (SISY)*
- Kojima R, Sugiyama O, Hoshiba K, Suzuki R, Nakadai K (2018) HARK-bird-box: a portable real-time bird song scene analysis system. In: *Proceedings of 2018 IEEE/RSJ international conference on intelligent robots and systems (IROS)*. <https://doi.org/10.1109/IROS.2018.8594070>
- Küçüktopcu O, Masazade E, Ünsalan C, Varshney PK (2019) A real-time bird sound recognition system using a low-cost microcontroller. *Appl Acoust* 148:194–201. <https://doi.org/10.1016/j.apacoust.2018.12.028>
- Navine AK, Denton T, Weldy MJ, Hart PJ (2024) All thresholds barred: direct estimation of call density in bioacoustic data. *Front Bird Sci* 3:1380636. <https://doi.org/10.3389/fbirs.2024.1380636>
- Opaev A (2016) Relationships between repertoire size and organization of song bouts in the Grey-crowned Warbler (*Seicercus tephrocephalus*). *J Ornithol* 157:949–960. <https://doi.org/10.1007/s10336-016-1342-6>
- Pérez-Granados C (2023) BirdNET: applications, performance, pitfalls and future opportunities. *Ibis* 165:1068–1075. <https://doi.org/10.1111/ibi.13193>
- Stowell D, Wood MD, Pamuła H, Stylianou Y, Glotin H (2019) Automatic acoustic detection of birds through deep learning: the first bird audio detection challenge. *Methods Ecol Evol* 10:368–380. <https://doi.org/10.1111/2041-210X.13103>
- Symes L, Sugai LSM, Gottesman B, Pitzrick M, Wood C (2023) Acoustic analysis with BirdNET and (almost) no coding: practical instructions. Zenodo. <https://doi.org/10.5281/zenodo.8357176>
- Tseng Y-C, Eskelson BNI, Martin K, LeMay V (2021) Automatic bird sound detection: logistic regression based acoustic occupancy model. *Bioacoustics* 30:324–340. <https://doi.org/10.1080/09524622.2020.1730241>
- Ware L, Mahon CL, McLeod L, Jetté J-F (2023) Artificial intelligence (BirdNET) supplements manual methods to maximize bird species richness from acoustic data sets generated from regional monitoring. *Can J Zool* 101:1031–1051. <https://doi.org/10.1139/cjz-2023-0044>
- Weiss M, Hultsch H, Adam I, Scharff C, Kipper S (2014) The use of network analysis to study complex animal communication systems: a study on Nightingale song. *Proc Biol Sci* 281:20140460. <https://doi.org/10.1098/rspb.2014.0460>
- Wheelhouse LM, Hodder DP, Otter KA (2022) The retention of non-commercial hardwoods in mixed stands maintains higher avian biodiversity than clear-cutting. *J for Res* 95:572–581. <https://doi.org/10.1093/forestry/cpac012>
- Wood CM, Kahl S (2024) Guidelines for appropriate use of BirdNET scores and other detector outputs. *J Ornithol* 165:778–782. <https://doi.org/10.1007/s10336-024-02144-5>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.